



Use of Fiber Reinforced Polymer Composites as Reinforcing Material for Concrete

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Abstract: Although the potential of continuous fiber reinforced polymers (FRPs) was recognized more than 50 years ago, bottlenecks such as vulnerability to static fatigue, ultraviolet radiation, alkaline environment, etc., continued to restrict their use as a construction material. Extensive research across the world during approximately the last 25 years has led to a better understanding of properties and behavior of the FRPs under different conditions. Given their lightweight, noncorrosive, and nonmagnetic nature and their high tensile strength, FRP composites are being used as reinforcement in concrete, especially in prestressed concrete construction, anchors for slope stabilization, and use in special structures such as high-speed linear motor railway tracks, MRI units of hospitals, and repair and rehabilitation works. An attempt has been made in this paper to comprehensively put together the various aspects of FRPs, including their structural and durability characteristics, the ongoing research in the areas related to their utilization as a reinforcing material for concrete structures, and the application of FRP sheets in rehabilitation of deteriorated concrete structures. Though issues related to the manufacture of FRP composites and design procedures have not been considered in detail, they have been included for information and completeness. The paper also presents the current thinking on the design and construction procedures using FRP material in conjunction with concrete and provides a brief discussion of some of the documented applications of FRP composites—rods and sheets—as reinforcement in concrete construction. It may also be mentioned that the paper is largely based on the case studies and research work carried out in Japan; appropriate references to related work in other parts of the world are also included, though in principle there is very little difference in the approaches being adopted across the world.

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Introduction

Continuous fiber reinforced polymer (FRP) composites have been used in automobile, electronics, and aerospace engineering for several decades, but their application in concrete engineering as a reinforcing material is relatively recent in origin. Besides the advances in the field of development of new fibers, etc., several developments in the construction industry have accelerated the efforts to apply FRPs as a reinforcing material in concrete. One of

the chief causes is the realization that the reinforcing steel, both in reinforced and prestressed concrete construction, is corrosion prone, though the malady may take more than 20–30 years to show symptoms! This realization obviously prompts the desire to use noncorrosive materials such as FRP, especially in environments where steel has been shown to be vulnerable.

Fig. 1 is a representation of how the predominant construction material has changed in the last about 7,000 years. It can be seen the FRPs are the youngest in the family of civil engineering construction materials. Extensive research across the world during the last 25 years or so has led to a better understanding of the properties and behavior of the FRPs under different conditions, and more extensive use of FRPs is likely to be seen in the coming years (e.g., Taerwe et al. 2001). It may be pointed out that the application of FRPs in other disciplines is mostly in the form of essentially 2D thin-shell or plate-like elements, where buckling of the elements is the main design criteria. On the other hand, the use of FRPs as a reinforcing material in concrete construction largely focuses on utilization of FRPs arranged in a single direction and subjected to high levels of tensile load. In other words, concrete engineering applications of FRPs need a more careful consideration of such properties as creep, fatigue, and modulus of fracture.

Besides being noncorrosive, the FRPs have a much better strength-to-weight ratio than steel and this makes them an ideal material for applications in repair, rehabilitation, and strengthening works. Also, in certain special applications (e.g., high-speed

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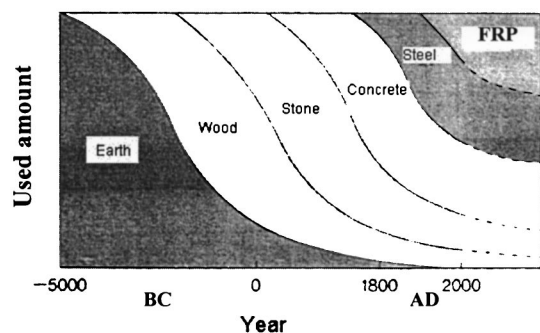


Fig. 1. Dominant construction materials in different ages

linear motor railway tracks in Japan) use of a nonmagnetic material such as FRPs has an added advantage. Though research in several areas pertaining to the utilization of the FRPs as a reinforcing material is still ongoing, engineers have used a blend of caution and judgment to proceed with the development of application-related technology, and several cases of actual application have been reported (e.g., Jesse and Curbach 2001; ACC 1995, 1998).

An effort has been made in this paper to provide an overview to the properties and behavior of commercially available FRPs and to briefly discuss the application of FRPs as a reinforcing material in reinforced and prestressed concrete construction. Some details of the application of continuous FRP sheets, especially in the area of retrofitting and strengthening of concrete structures, have also been presented. This study presents the current thinking on the design and construction procedures using FRP material in conjunction with concrete. Although most of the examples cited in the paper are drawn from the developments in Japan, the basic trends and overall approach to the use of FRP as a reinforcing material is the same in other parts of the world.

Historical Notes

Given the high strength and lightweight nature of the FRPs, their potential as a reinforcing material in the prestressed concrete construction was realized early; it was the subject of research in the U.S., the erstwhile Soviet Union, and the U.K. as early as the 50s and 60s (Kobayashi et al. 1987). In the tests carried out at that time, it was found that though a tensile strength of as much as $3,620 \text{ kgf/mm}^2$ was observed using a $3\mu\text{m}$ monofilament, the strength of the rods or other composite materials was much lower (about 700 kgf/mm^2). Besides this reduction in the load-carrying capacity when made into a composite material, it was found that the FRPs were susceptible to premature failure under sustained load. It was estimated that if the load was applied over a long period of time, the rods could fail at load levels as low as 60–70% of the tensile strength observed in the laboratory. Thus, assuming that a maximum load of about 70% of the tensile capacity is applied at the time of prestressing, the actual prestressing load is only about 45–50% of the tensile strength observed in the laboratory. Even so, it must be borne in mind that a permissible load level of even 400 kgf/mm^2 was higher than the levels of 160 kgf/mm^2 permissible in steel at that time.

Premature failures in the FRPs were attributed to the phenomenon of static fatigue, which resulted from the irregular distribution of tensile stresses among the fibers in an FRP composite.

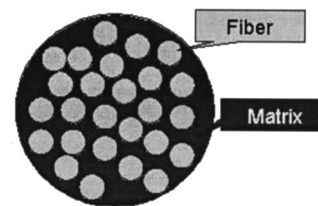


Fig. 2. Schematic representation of cross section of continuous fiber reinforced polymer composite

Such a distribution of stresses resulted from the fact that all the fibers were not properly aligned and thus did not fully participate in carrying the applied tensile load. The problem persisted till the process of pultrusion for the manufacture of FRP products was developed in the 70s (Kobayashi et al. 1987).

German scientists used FRP products using glass fiber and polyester resin as a matrix to build an experimental bridge with a span of only 6.5 m in the early 80s (Waaser and Wolf 1986). The tensile strength and the modulus of elasticity of the FRP used were 150 kgf/mm^2 and $5,100 \text{ kgf/mm}^2$, respectively. Over the last 20 years, substantial progress has been made in all areas, including manufacture, quality control, and applications. Some of the more important features are discussed in this paper.

Besides overcoming the problem of static fatigue, persistent research efforts were required to overcome such problems as the durability of glass fibers in the highly alkaline environment of concrete (Uomoto 2001) and the rapid deterioration of epoxy resins upon exposure to ultraviolet radiation (Yamaguchi 1998). Development of appropriate test methods and standards for the FRP composites also placed a major challenge to the professional institutions and researchers. It may be noted that comprehensive specifications for the design and construction of concrete structures using FRPs have been published only recently (JCI 1998; ACI 2001; JSCE 1997, 2001).

Continuous Fiber Reinforced Polymer Materials

General

As against short fibers of steel that are sometimes used to improve the tensile strength and postcracking behavior of the brittle concrete matrix, continuous long fibers can be used as a replacement of the steel as a reinforcing material for reinforced and prestressed concrete construction.

From the schematic representation of the cross section of continuous FRP composites shown in Fig. 2, it is clear that they can be seen as made up of extremely fine fibers embedded in a matrix. The volume fraction of the fibers in commonly available FRPs ranges from about 50 to 65%. As discussed in the following paragraphs, the properties of FRP composite products are related to (1) the properties of the fibers used, (2) the properties of the matrix used, and (3) the volume fraction of the fibers.

Constituent Fibers

Fibers made from carbon, aramid, glass, and polyvinyl alcohol are commonly used for the manufacture of FRP. Table 1 gives a brief overview of the properties of the commonly used fibers in FRPs used in the construction industry. As can be seen, the specific gravity of the fibers ranges between 1.4 for aramid fibers to

Table 1. Properties of Fibers used in Fiber Reinforced Polymer Composites

Item	Carbon fiber				Aramid fiber		Glass fiber		Polyvinyl alcohol fiber (high strength)
	Polyacrylic Nitril Carbon		Pitch Carbon						
	HIGH STRENGTH	HIGH YOUNG'S MODULUS	ORDINARY	HIGH YOUNG'S MODULUS	Kevlar 49 Twaron	Tech-nora	E-Glass	Alkali-Resistant Glass	
Tensile strength (MPa)	3430	2,450–3,920	764–980	2,940–3,430	2,744	3,430	343 0–3 528	1,764–3 430	2254
Young's modulus (GPa)	196–235	343–637	37–39	392–784	127	72.5	72.5–73.5	68.6–7	59.8
Elongatio n (%)	1.3–1.8	0.4–0.8	2.1–2.5	0.4–1.5	2.3	4.6	4.8	4.5 2–3	5.0
Density (gm/cm ³)	1.7–1.8	1.8–2.0	1.6–1.7	1.9–2.1	1.45	1.39	2.6	2.27	1.30
Diameter (μm)	5–8		9–18		12		8–12		14

about 2.6 for glass fibers. The diameters of the fibers range from about 6 microns for carbon fibers to about 15 microns in the case of aramid and glass fibers.

Fig. 3 gives a diagrammatic representation of the stress-strain curves for some commonly used FRP composites. Curves for steel and PC tendon have also been included in the figure only for reference. It can be seen that though all the fibers exhibit a higher tensile strength than steel, the elongation before the material finally fails is very small for the fibers. In fact, it is this lack of extensibility before failure that often makes the FRP composites unusable as normal reinforcement in RC structures, where large deformations before collapse are a design requirement. The carbon fibers have a higher modulus of elasticity and a lower elongation as compared with others. This reduces the toughness of the carbon FRPs and also makes carbon fiber products vulnerable to impact loads. The polyvinyl alcohol fiber is light and has better extensibility, but it rather belongs with other fibers with a relatively low tensile strength and Young's modulus. The following paragraphs give a brief description of the properties of the individual fibers (CCC 1998).

Carbon Fiber

These are made from either petroleum or coal pitch and polyacrylic nitril (PAN). Each fiber is an aggregate of imperfect fine

graphite crystals, and its characteristics vary depending on the composition and the orientation of the crystals. Commercially available carbon fibers are usually classified into pitch carbon and PAN carbon fibers, the pitch carbon being ordinary or high Young's modulus fiber; the modulus of elasticity for the ordinary fiber of the pitch carbon family is about 20% of those of the PAN family. High Young's modulus fibers, on the other hand, provide an identical or higher Young's modulus than the fibers of PAN family.

Aramid Fiber

This is a highly oriented organic fiber, and three types of these fibers are available—Kevlar, Twaron, and Technora—which are essentially trade names. Kevlar and Twaron are aromatic polyamide fibers that have their benzene nuclei bonded linearly by aramid. Technora, on the other hand, is an aromatic polyetheramide fiber whose nuclei are bonded by ether.

Glass Fiber

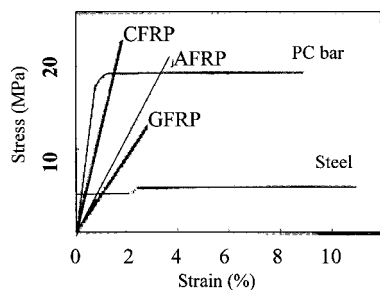
Two types of glass fibers are commercially available for use—E-glass fiber and alkali-resistant glass fiber. The former contains large amounts of boric acid and aluminate, while the latter contains a considerable amount of zirconia, which serves to prevent corrosion by alkali attacks from cement matrix.

Polyvinyl Alcohol Fiber

As the name suggests, these are manufactured using polyvinyl alcohol of a high degree of polymerization. The fiber is rolled during the manufacturing process to provide added strength and elasticity.

Mixed Fibers

Recent years have also seen the introduction of products, where fibers of different origins are combined to produce a final product having desired properties. Nishimura reported the development of a combined fiber "hybrid" with glass fibers in the middle, surrounded by aramid fibers (Nishimura et al. 1996). Fig. 4 shows a schematic representation of the hybrid FRP composite. Such a structure provides the glass fibers extra protection in the alkaline

**Fig. 3.** Typical stress-strain curves for some fiber reinforced polymer (FRP) composites

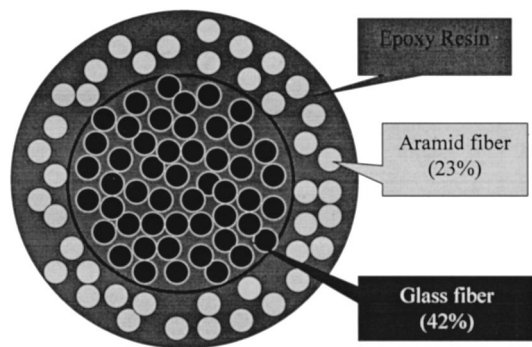


Fig. 4. Schematic representation of hybrid fiber reinforced polymer composite

environment of the concrete. NEFMAC, which stands for “new fiber composite material for concrete,” is another example of such a material that uses continuous carbon and glass fibers impregnated with resin and formed into a lattice, as schematically given in Fig. 5.

Resins

Though there is limited evidence of use of special cement or silica fume, most of the FRPs use various kinds of resins, including epoxies, vinyl esters, and polyesters (Fig. 2). The specific gravity of the resins used varies between 1.1 and 1.4. Although the resin does not carry any tensile load directly, it acts as a filler material and holds the fibers together. The resins also play a crucial role in transferring the load and protecting the fibers in much the same manner as the surrounding concrete protects the embedded reinforcing bars. For example, the durability of glass-fiber-reinforced composites can be substantially increased if individual fibers are coated with a very thin layer of resins.

It is important that the fracture strain of the resins and the compatibility with the fibers themselves are taken into consideration in an appropriate manner. For example, epoxy resins have a low strain capacity and work well in conjunction with carbon fibers, which also have low extensibility, but the results are not so encouraging when epoxy resins are used with aramid or glass fibers, which have larger strains at failure. Vinyl ester formulations serve as better “filler” products in such an instance.

Substantial lowering in the tensile strength of the FRP rods has been reported in cases when the impregnation with the resin is improper. This lowering of the tensile strength can be attributed to stress concentrations in certain fibers as a result of the uneven

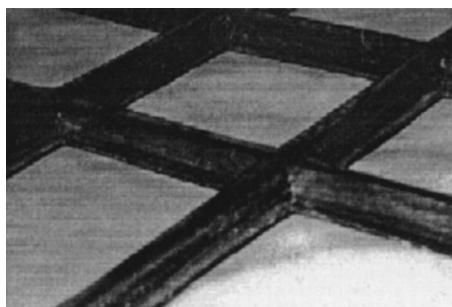


Fig. 5. Schematic representation of NEFMAC

distribution of applied stress arising from the improper impregnation of the matrix (Hodhod 1992).

In the case of application of continuous FRP sheets, the resins are usually applied in situ while wrapping or attaching the sheet, and care needs to be taken to ensure that hardening and strength development proceed at a satisfactory rate.

Manufacture of Fiber Reinforced Polymers

As mentioned above, earlier attempts to utilize FRPs failed primarily on account of manufacturing difficulties, which led to static-fatigue-induced failure in the rods. Most of the present-day manufacture of FRP products used for reinforcement is by “pultrusion,” a continuous process that combines pulling and extrusion for manufacturing composite sections that have the same cross section and shape. The process consists of pulling a fiber material through a resin bath and then through a heated shaping die, where the resin is cured. Important steps of the process and other methods of manufacture have been described in the literature (e.g., Mallick 1988; Agarwal and Broutman 1990).

The applied tension on the fibers during the manufacture eliminates twists and straightens them. When such products are actually stressed while testing or in a real application, a more uniform distribution of stresses among all the fibers is achieved, thus preventing static fatigue (at low load levels). It may be pointed out that the tension applied at time of manufacture is small, so as not to impair the performance of the composite as a construction material.

As far as the production of 2D and 3D FRP products is concerned, the grids are manufactured by impregnating bundles of fibers with resins and alternately laying them in *X* and *Y* directions under tension and pressure before shaping them into grids. The three-dimensional textiles are manufactured either by alternately laying bundles of fibers in *X*, *Y*, and *Z* directions in several layers while impregnating them with resins or by impregnating fibers already shaped into textiles with resins. Fibers woven into textiles using special cement or silica fume are produced as flat or curved plates.

FRP sheets are another variety of FRP materials available for use, especially in jobs related to the reinforcement and strengthening of existing structures (Iwahashi et al. 1996; Toritani et al. 1997; Yagishita et al. 1997). These sheets consist essentially of fibers laid out in one or two directions and held in place by a light adhesive. They can be classified as (1) one-way textile, (2) two-way textile, or (3) one-way pre-impregnated textiles. The resin and hardeners are applied to the sheets at the time of actual application, as discussed later in this paper (JCI 1998).

Properties of Fiber Reinforced Polymer Products

General

Mechanical properties such as the strength and modulus of elasticity of a product are related to the properties and proportions of the constituent materials. A laboratory study has demonstrated the possible use of the rule of mixtures, to estimate the properties of the FRP rods from the properties and the volume fraction of the parent fibers (Uomoto and Nishimura 1994). The study used the following equations:

$$E_{FRP} = E_1 V_f + E_2 (1 - V_f) \quad (1)$$

$$\sigma_{FRP} = \sigma_1 V_f + \sigma_2 (1 - V_f) \quad (2)$$

Table 2. Comparison of Calculated and Experimental Values of Tensile Strength using Rule of Mixtures

V_f	Item	AFRP	GFRP	CFRP
45	Experimental (1)	134.5	140.1	124.4
	Calculated (2)	175.1	113.0	150.8
	(1)/(2)	0.77	1.24	0.83
55	Experimental (1)	168.9	168.9	133.5
	Calculated (2)	214.0	138.1	184.3
	(1)/(2)	0.79	1.22	0.72
66	Experimental (1)	204.2	176.9	148.4
	Calculated (2)	256.7	165.7	221.1
	(1)/(2)	0.80	1.07	0.67* (0.81)

As an approximation, if the contributions of the matrix phase are neglected, the above equations can be written as follows:

$$E_{FRP} = E_1 V_f \quad (3)$$

$$\sigma_{FRP} = \sigma_1 V_f \quad (4)$$

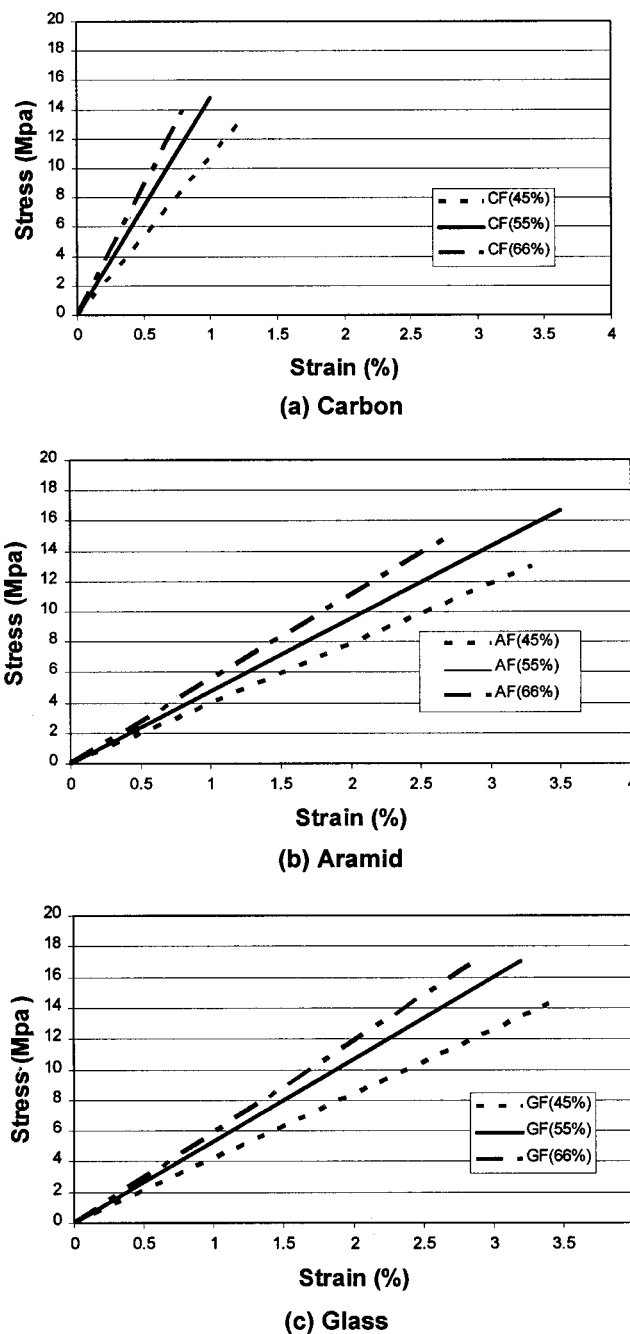
Table 2 shows the variation in the values of the estimated and observed values of the tensile strength and modulus of elasticity, assuming that all the tensile load is carried only by the fibers. It can be seen that the two are in reasonable agreement. The following paragraphs discuss briefly some of the important aspects of properties of the FRP, which are relevant to their application as a construction material.

Tensile Strength and Modulus of Elasticity

Given the fact that FRPs are used as tensile reinforcement in concrete construction, it is important that the behavior of the FRPs under tension is accurately understood. An extensive study was undertaken by Uomoto to better understand the variation in these properties with the type and content of the fibers and the statistical variation therein (Uomoto and Nishimura 1994). 100 specimens were tested at each test condition, and Table 3 gives the representation of the results obtained, including the statistical

Table 3. Tensile Strength and Modulus of Elasticity of Fiber Reinforced Polymer Rods

Parameter	V_f	Fiber type	AFRP	GFRP	CFRP
Strength (kgf/mm^2)	45	Mean Value	135	140	124
		SD	9.49	5.83	11.8
		COV	0.0703	0.0416	0.0952
	55	Mean Value	169	169	134
		SD	14.8	8.53	11.2
		COV	0.0876	0.0505	0.0836
	66	Mean Value	204	177	148
		SD	5.73	12.2	27.0
		COV	0.0281	0.0689	0.1824
Elastic modulus (kgf/mm^2)	45	Mean Value	3748	4274	11202
		SD	216.1	52.2	259.5
		COV	0.0576	0.0122	0.0232
	55	Mean Value	4570	5211	13530
		SD	194.5	47.6	236.3
		COV	0.0426	0.0091	0.0175
	66	Mean Value	5460	6024	15714
		SD	241.2	101.2	379.4
		COV	0.0442	0.0168	0.0241

**Fig. 6.** Typical stress-strain curves obtained using fiber reinforced polymer rods

parameters, such as mean and standard deviation. Typical stress-strain curves obtained using rods having different fibers and fiber volumes are given in Fig. 6. Given the fact that the properties depend on various factors such as the type and content of the fibers and the type of the resin, it is difficult to make generalized conclusions about the properties of the bars. However, the following are generally known principles that guide the present thinking on application of FRP as a construction material.

- The stress-strain behavior of all FRP materials is essentially linear till the point of failure, which is sudden;
- The fracture strains are much smaller when compared with steel and hence the material (FRP) should be used with great caution for applications when large deformations or ductility in the member is required (for example, in flexural failure);

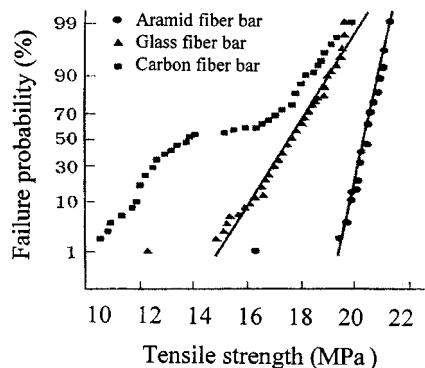


Fig. 7. Representation of tensile strength of fiber reinforced polymer rods in terms of cumulative probability versus stress

- The modulus of elasticity is about half that of steel (except in some cases of carbon FRPs); and
 - The tensile strength of FRPs is much higher than that of steel.
- It may also be mentioned that the properties of the FRP may not necessarily follow a “normal distribution,” research effort has been directed to apply different statistical distributions and represent the failure strength in terms of a cumulative probability distribution (Hodhod 1992). A sample of such a representation is shown in Fig. 7.

Fig. 8 is a representation of typical failure patterns observed in the FRP rods and strip, depending upon the volume content of the fibers and the properties of the matrix. It can be clearly seen that FRP composites show several modes of failure besides the rupture of the fibers—these include cracking of the matrix, partial interfacial adhesion failure, debonding between the fiber and the matrix, and longitudinal cracking. Studies using acoustic emission to better understand the failure of FRP also corroborate such a hypothesis (Uomoto et al. 1992). It has also been pointed out that aramid and glass fibers show consistency in the mode of failure regardless of the failure load level. However, the carbon FRPs fail in different modes, including fiber pull-out from the matrix and simultaneous failure at the anchorage, fan-type failure, and failure containing components of both the above (Uomoto and Nishimura 1994).

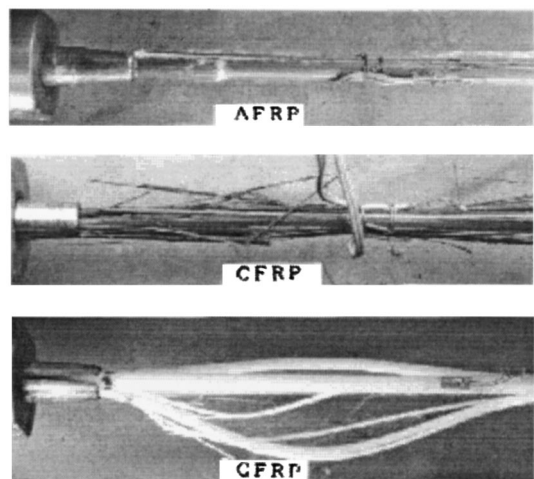


Fig. 8. Representation of typical patterns of tensile failure of fiber reinforced polymer rods

Fatigue Strength, Creep and Relaxation

As mentioned above, static fatigue was one of the major impediments in the application of the FRPs in concrete structures. Thus a substantial amount of research work has been carried out to better understand the increase in the strain at constant load (creep) and the decrease in the load at constant strain (relaxation) behavior of the FRPs (e.g., Uomoto 1995, 1998; Saadatmanesh and Tannous 1999). It has been pointed out that the extent of creep deformation observed in FRP composites is almost twice that observed in steel. However, the lower value of the modulus of elasticity can be used to an advantage when losses are to be accounted for (as in the case of prestressed concrete), as the loss of a certain amount of deformation leads to a lesser extent of force reduction in case FRP reinforcement is used.

Compressive Strength

Because of the inherent difficulty in testing FRP specimens in compression because of their tendency to buckle, there is only limited information available on the subject (Mallick 1988; Piggett and Harris 1980). It has been reported that the compressive strength and the modulus of elasticity of Kevlar fibers are significantly lower in compression than the corresponding values in tension. In the case of carbon and glass fibers, the difference is much less, and there is virtually no difference in the values obtained for boron-reinforced FRP composites. Given the scarcity of the information, it remains a challenge to adapt published data to the analysis and design procedures, as far as the application of FRPs to a reinforcing material in concrete is concerned.

Bond with Concrete

Whereas excellent bond between the prestressing reinforcement and the surrounding concrete is a requirement in normal reinforced and pretensioned prestressed concrete construction, the property is only of marginal significance in the case of posttensioned construction. Since there is an apprehension of inadequate bond between the concrete and the FRP reinforcement, most of the application of FRPs has been reported in posttensioned prestressed concrete construction. However, if special measures like braiding are taken, it has been shown that adequate bond can indeed be ensured (JSCE 1997). It has also been found that immersion in water tends to increase the bond between the surrounding concrete and the FRP reinforcement due to expansion of the polymers on account of reaction with water (Katz et al. 2001). Eq. (5) has been proposed to estimate the development length, which may be taken to be a measure of the bond strength (JSCE 1997):

$$f_{\text{bond}} = \frac{0.28\alpha f_{ck}^{2/3}}{\gamma_c} \quad (5)$$

In the above expression, α is a modification factor in the estimation of the bond strength for continuous FRPs. The JSCE recommends a value of 1.0 in cases where the bond strength is equal to or greater than that of the deformed bars. In other cases, a lower value is to be determined on the basis of experimental results. Fig. 9 shows a comparison between the actual and estimated values of the bond strength. It can be seen that the equation used estimates the bond strength reasonably well.

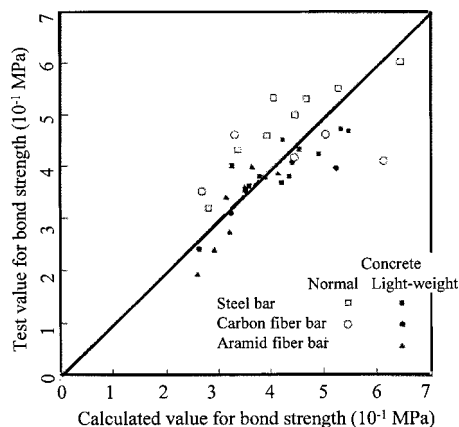


Fig. 9. Comparison of estimated and actual bond strength between fiber reinforced polymer rods and concrete

Durability

In cases when the FRPs are used within normal concrete, they need to show high durability in an alkaline environment, whereas if they are used for external prestressing, durability against UV radiation could be an additional requirement. Results of experimental research in the area of durability characteristics of FRP materials have been published by several researchers (Katsuki 1997; Yamaguchi 1998; Uomoto 2001). Tests were carried out using fibers *per se* and FRP rods to study the static fatigue fracture, fatigue fracture, alkali resistance, ultraviolet ray resistance, freeze-thawing resistance, high temperature and fire resistance. The results summarized in Table 4 indicate that

- Carbon fibers and FRP rods have good durability characteristics.
- Aramid fibers and FRP rods show good properties except under static fatigue, UV radiation, and acidic environment.
- Glass fibers may show poor durability characteristics as far as their alkali resistance is concerned, though they were found to be satisfactory in the acidic and freeze-thaw environment.
- FRP materials in general show poor performance at high temperatures, and therefore their use should be avoided when fire resistance is a requirement.
- Sheets made with carbon fibers need to be checked for durability against UV radiation.

Table 4. Durability of Fibers and Fiber Reinforced Polymer Rods

Particulars	Fiber			Notes
	Carbon	Aramid	Glass	
Alkali resistance	95%	92%	15%	NaOH, 40°C, 1,000 h
Acidic resistance	100%	60–85%	100%	HCl, 40°C, 120 days
Ultraviolet ray resistance	100%	45%	81%	0.2 MJ/m ² /h, 1,000 h
Particulars	FRP rods			Notes
	CFRP	AFRP	GFRP	
Static fatigue	91%	46%	30%	20°C, 100 years (calculated)
Cyclic fatigue	85%	70%	23%	100 MPa Amp., 2 million cycles
Alkali resistance	100%	98%	29%	NaOH, 40°C, 120 days
Ultra-violet ray resistance	77%	69%	90%	3 years exposure
Freeze-thaw resistance	100%	100%	100%	–20 to +15°C, 300 cycles
High temperature resistance	80%	75%	80%	–10 to 60°
Fire resistance	75%	65%	75%	350°C

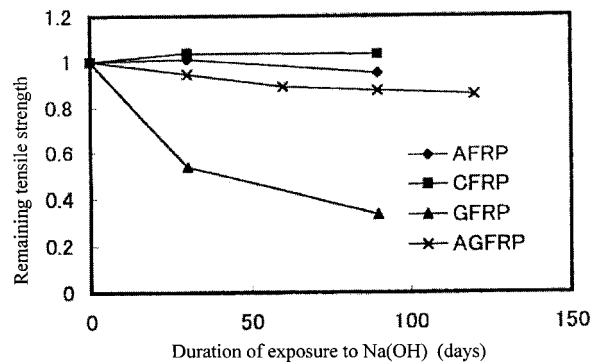


Fig. 10. Variation of observed tensile strength with continued exposure to sodium hydroxide

- There is also a need to limit the tensile load depending on the duration of the load, in cases when the FRPs are used as an internal reinforcement.

Fig. 10 shows the variation of the tensile strength observed for the different FRP rods under the continued action of sodium hydroxide. It can be seen that the glass fibers are expectedly most vulnerable. Also, the performance of the hybrid fiber shown in Fig. 4 clearly shows the effectiveness of the aramid fibers in protecting the interior core of glass fibers.

It may be mentioned here, however, that the long-term durability of the FRP products is related to various factors, such as the properties of the resin, fiber, interfacial characteristics, fabrication process, and the environment. Thus, the results of accelerated and other tests carried out in the laboratory are only representative in nature, and an appropriate test program taking into account the actual usage should be developed, more or less on a case-to-case basis. Some of the standards that have been developed in Japan for testing of FRP rods and sheets have been listed in the appendixes of this paper.

Commercially Available Fiber Reinforced Polymer Composites

Though it is, in principle, possible to produce FRP products (cable, rods, flat bars, tendons, grids, and fabrics) with a great

Table 5. Commonly Available Fiber Reinforced Polymer Composites for Application as Reinforcement

Material	Type of FRP	Diameter (mm)	Tensile strength (MPa)	Elongation at rupture (%)	Young's modulus (GPa)	Prestressing method	Introduced prestress
Aramid	Round bar	6.1	1,744	3.66	4.8	Compression friction type	Tensile strength Stress of 4.9 MPa at
	Round bar	6	1,911	3.2	5.7	Compression friction type	0.31, 0.4
	Braided bar with and without sand	8	1,421	2.2	6.5	Pretension	0.2, 0.4, 0.6
	Braided bar with sand	9, 11, 13.5, 15.5, 17.5	1,215–1,382	1.92–2.3	5.9–6.5	Pretension	0.29, 0.32,
						Posttension	0.47, 0.50
	Eight round bars	6	1,921	4.0	4.8	Posttension	0.5
	Rope	8.3	1,921	2.5	7.8	Pretension	0.3
	Deformed bar	6	1,862	3.6	5.3	Posttension	0.49
	Deformed bar round bar	8	1,431	1.0	14.3	Pretension	Pretension 0.2
						Block beam	Block 0.6
Carbon	Twisted seven FRP bars	9.8	2,068	1.5	13.7	Pretension	0.13–0.19
	Bars put in sand	9.5	1,617		14.8	Pretension	0.17–0.50
	Twisted seven simple	5, 6	1,862	1.3	13.7	Pretension	0.4, 0.55, 0.7
	Twisted seven FRP bars	12.1	1,803	1.32	13.7	Pretension	0.55
	Twisted seven FRP bars	6	1,548	1.3	11.9	Pretension	0.4
	Eight round bars	8×8 dia.	1,803	1.21	14.7	Posttension	0.5, 0.65
	Eight round bars	8×8 dia.	1,803	1.2	13.9	Posttension	0.47
	Twisted 7 or 19 CFRP bars	12.5, 17.5, 25.5	2,068, 1,931, 1,911	1.5	14.2, 13.6, 12.9	Posttension	0.27, 0.44, 0.51, 0.54, 0.75
Glass	Grid		657	2.18	3.1	Pretension	0.37, 0.41, 0.5
	Deformed bars	6, 8	1,205, 1,313	3.1–3.3	4.0–4.3	Pretension	0.27

deal of flexibility depending upon the requirements, some of the more commonly available products are listed in Table 5 along with their basis properties. In addition to the above products, FRP sheets are also available, which are being increasingly used for the repair and reinforcement of concrete structures.

FRP materials for use as reinforcing materials in concrete construction are available in different shapes. Fig. 11 shows a sample of some of the common shapes of FRP materials used in the construction industry—grid, spiral rope, braided rope, deformed bar, round bar, and flat bar. It can be seen from the figure that special efforts are made at times to improve the bond between the FRP reinforcement and the surrounding concrete or the grout. Such efforts include braiding of fibers or giving some kind of special surface treatments. Fibers may also be wound around the surface in a twill or spiral pattern or may be coated with sand. It may also be pointed out that the 3D FRP products are rarely used, as they pose a major problem in their transportability.

Testing Fiber Reinforced Polymer Composites

Given the diversity of the products available and the relatively short history of the use of FRP materials in the construction industry, it is only natural that the relevant test methods and specifications are still in their infancy. Some of the standards that have been developed in Japan for testing of FRP rods and sheets have been listed in the Appendixes 1 and 2 of this paper. Only one aspect of properly holding the composite at the ends has been taken up for a brief discussion here.

Fig. 12(a) shows an FRP being held in a specially designed V-shaped split chuck for tensile testing. It has been found that

unless the restraining compressive load is applied uniformly along the entire surface held within the chuck, the FRP could rupture on account of split tension or stress concentrations, even before the fibers reach their full load-carrying capacity (Uomoto and Nishimura 1994). In fact, the existence of two clear peaks in the probable failure loads of carbon fibers has been clearly demonstrated (Uomoto and Nishimura 1994). Two possible approaches have emerged to achieve the desired objective in this regard:

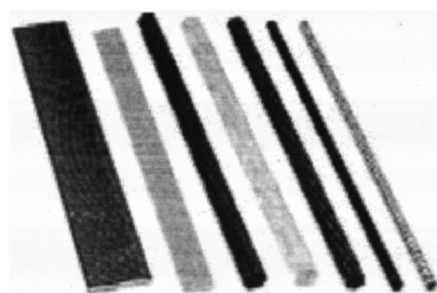
1. Shape the interior of the split chuck appropriately in terms of the end angle and the cross section of the V; and
2. Embed the FRP in a threaded pipe using an appropriate expansive agent. The threads are then used to connect to the chuck and in applying the tensile load. Fig. 12(b) shows an example of a holding device based on this principle.

Use of Fiber Composites in Concrete Engineering

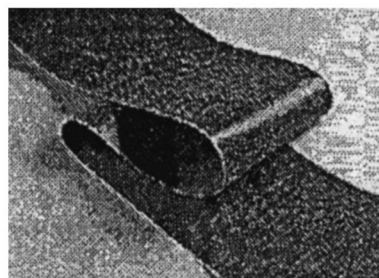
General

Some of the basic properties of the available FRP materials have been discussed above. In the last several years, considerable research effort has also been directed to study the possible application of FRP as a reinforcing material in concrete construction. There are several aspects to the research work; the following broad classification has been made to facilitate discussion in this paper:

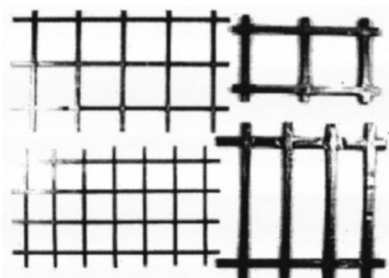
1. Application of FRP rods to the construction of new reinforced concrete structures;
2. Application of FRP rods to the construction of new prestressed concrete structures; and



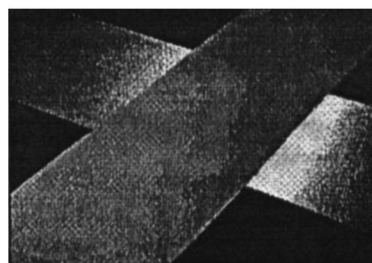
(a) One dimension



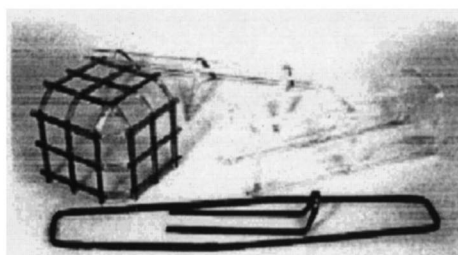
(d) One-dimensional sheet



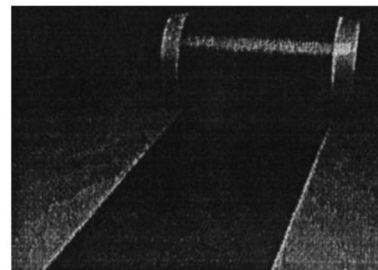
(b) Two dimensions



(e) Aramid fiber sheet and glass fiber sheet



(c) Three dimensions



(f) Carbon fiber sheet

Fig. 11. Common shapes of commercially available fiber reinforced polymer composites

3. Application of FRP sheets in repair and rehabilitation of different kinds of concrete structures

Whereas the following paragraphs discuss the application of FRP rods in concrete construction, the issues relating to the use of FRP sheets are taken up in the next section of this paper.

Application of Fiber Reinforced Polymer Rods in Construction of New RC Structures

Given the almost linear stress-strain behavior of FRP reinforcement, RC beams reinforced with FRP rods generally behave as continuous elastic bodies until cracking starts in concrete, and the behavior can be modeled using conventional models used for the steel-reinforced beams. Once the tensile stress in the bottom (concrete) fiber reaches the tensile strength of the concrete, and flexural cracking is initiated, the behavior is quite different. Concepts like that of “a balanced section” are equally valid in the case of an RC beam reinforced with FRP rods. In cases when FRP is used as a reinforcing material, care needs to be taken to define the concept, taking into account the variation of the fiber volumes, etc., in the reinforcement used.

Initial Flexural Cracking

It is well known that although the occurrence of flexural cracks in concrete is related to the strength of concrete, the distribution of

the cracks and the spacing is related to the bond between the reinforcing material and the surrounding concrete. It may be noted that if certain basic precautions are taken, the bond between the concrete and the FRP rods does not pose a major problem. In the case of grid-form FRP composites, flexural cracks occur in the vicinity of the intersection points of the FRP grids. It has also been reported that the maximum crack intervals is about two times the grid spacing, and that the width of the cracks is not as large as that observed in cases when steel bars are used and is nearly proportional to the elongation of the principal reinforcing materials.

Flexural Rigidity After Cracking

When flexural cracking starts, it is known that the deflections increase and the neutral axis in the cracked portions rises. Now, since the modulus of elasticity of FRP rods is considerably lower than that of steel (except for a few carbon-fiber-reinforced FRPs), the deflections and the extent of the rise of the neutral axis are considerably more than those observed for conventional steel bars.

Ultimate Strength

Basically speaking, the ultimate load-carrying capacity is determined by either the compression failure of the concrete or the

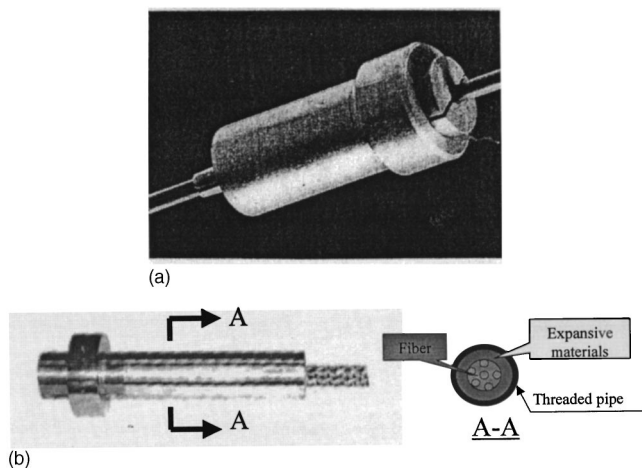


Fig. 12. (a) Typical V-shaped split chuck to hold fiber reinforced polymer composites during tensile testing; (b) special arrangement for holding fiber reinforced polymer composites during tensile testing

failure of the reinforcement in tension. When estimating the latter moment capacity, it needs to be borne in mind that the FRP materials do not have a very large fracture strain, and therefore the fracture is very brittle. In fact, if the reinforcement (FRP rods) is arranged in several layers, as it often happens for RC beams, it is likely that the capacity is decided as the extreme FRP fiber layer reaches fracture strain (even though the strands in others layers, i.e., those closer to the NA, are well below that strain level). In other words, in such cases, the full area of the FRP reinforcement is not effectively utilized. This is in contrast to the behavior seen in normal steel reinforcement, where because of the large deformability of the steel, the reinforcement continues to resist load, even after the extreme layer reaches “yield levels”—in other words, failure occurs only when the entire tensile reinforcement reaches yield level strains.

Application of Fiber Reinforced Polymer Rods in Construction of New PC Structures

In spite of lower modulus of elasticity, a relatively high strength, light weight, corrosion resistance, and low relaxation makes FRP a very suitable material for use as reinforcement in prestressed concrete construction. Continuous fibers have been effectively used in several prestressed concrete applications using both pre-tensioning and posttensioning methods. It may be noted that round bars are usually used only in posttensioned applications because of the lower bond that is achieved. In the absence of clear-cut design procedures and lack of understanding of the behavior of PC members, it is common to introduce prestressing up to the extent of about only 50% of the tensile strength of the FRP tendon. The following paragraphs briefly discuss the results from some of the research carried out in this direction.

Prestressing using Fiber Reinforced Polymers Tendons

Some of the issues involved in the tensile testing of the FRP composites discussed above in this paper are extremely relevant to the design of an appropriate system for prestressing the FRP composites. Fig. 13 shows a schematic representation of some of the holding chucks that are used at the time of prestressing FRP rods.

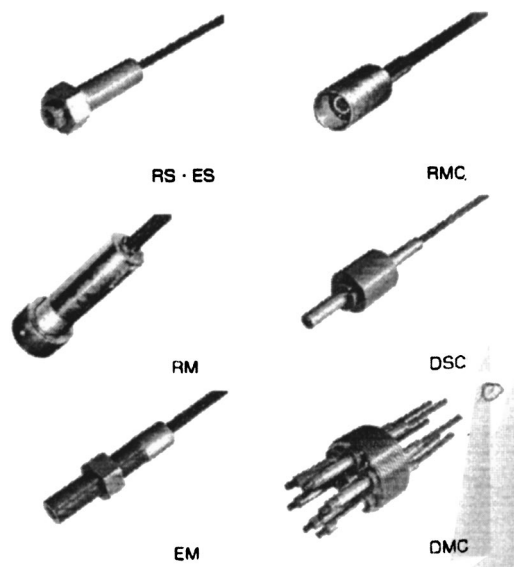


Fig. 13. Representation of some of the chucks used for prestressing fiber reinforced polymers tendons during construction

Load-Deformation Characteristics

Basically, the principles applicable to the prestressed concrete members using steel reinforcement are also applicable in case FRP tendons are used as reinforcement. It needs only to be borne in mind that, in the absence of a yield region in the stress-strain behavior of the FRP rods, the failure can only be either by the failure of the concrete (in compression) in the compression zone, or the breaking of the FRP rods.

Results from a study of prestressed concrete beams using FRP rods, and tested in flexure, have been reported. Several beams were tested varying the type of the fiber used, the level of the applied prestressing force, and the amount of fiber reinforcement (analogous to the different reinforcement ratio in conventional beams). It was found that the initial cracking load and the maximum strength increase as the introduced prestressing force and the volume of fibers in the aramid FRP are increased. A typical load-deflection diagram obtained using flexural members having FRP tendons is given in Fig. 14. It was further found from the tests that

1. The behavior before initiation of flexural cracking is linear and elastic. It may be pointed out that the values of deflection obtained match well with those estimated using conventional bending theory.

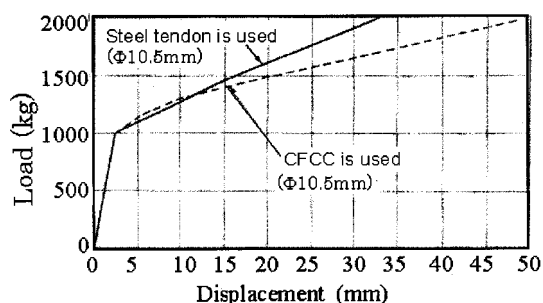


Fig. 14. Typical load deflection diagram for flexural member with prestressed fiber reinforced polymer tendons

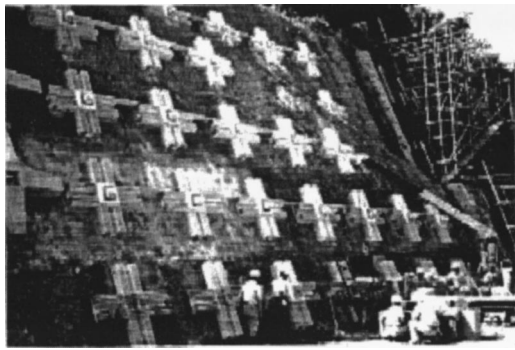


Fig. 15. Application of fiber reinforced polymer rods in soil stabilization

2. The values of deflection at loads higher than cracking loads can be predicted using conventional theories. Further, it should be pointed out that because of the lower values of the modulus of elasticity, the deflections in the case of FRP reinforcement are significantly higher than those in PC members reinforced with steel;
3. The failure load observed is in close agreement with the calculated value, estimated assuming complete bond between the FRP rods and the surrounding concrete. The failure is governed by the load-carrying capacity of the concrete (in compression), and is therefore sudden and brittle; and
4. Failure by compression of concrete occurs if the initial prestressing force is $0.6\sigma_u$ or less (σ_u : tensile strength of FRP) and the FRP are likely to rupture if the value is $0.6\sigma_u$ or more.

It has also been pointed out that in the case of unbonded members, the maximum load-carrying capacity of the member is about 60% of that of the bonded members.

Soil Stabilization

Fig. 15 shows application of prestressed FRP tendons in slope stabilization. The presence of the prestressing force in the anchor bolts increases the internal friction along the failure surface; this stabilizes the slopes and allows more steep cuts than can be made without the use of such bolts. It can be seen that because of the noncorrosive nature of the FRP rods, there is hardly a need to use protective cement mortar, which is an essential component in the schemes using steel reinforcement. Thus the holes used can be much smaller and the exercise becomes more economic and efficient.

Use of Fiber Reinforced Polymer Sheets

General

Besides the use of FRP rods and strips, as described above, continuous FRP sheets weighing between 200 and 410 g/m² have also evolved as a major application of FRP materials in the construction industry, especially in the repair and rehabilitation of deteriorated concrete structures (JCI 1997; JSCE 2001; ACI 2001). Application of steel plates for the rehabilitation of deteriorated structures has the basic drawback that the plate tends to separate from the concrete surface and may not thus behave in a manner monolithic with the concrete member. Also, it is difficult to ensure proper adhesion because of the uneven nature of the concrete surface. The development of FRP sheets addresses both

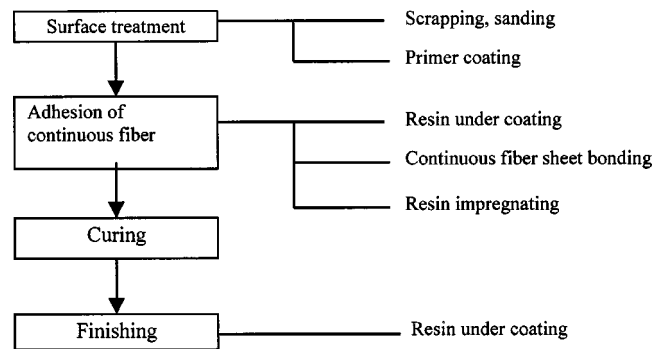


Fig. 16. Depiction of different steps in fiber reinforced polymer sheet application

of these issues effectively as the sheets adhere closely to the concrete surface and no special effort needs to be made to ensure a proper contact between the sheet and the concrete surface. Fig. 16 gives a brief description of the various steps in the operation, which can be summarized as follows:

1. Surface preparation, including impregnation of cracks and application of smoothing agents, if required, followed by primer coats;
2. Application of FRP sheets along with impregnating resins and hardeners. The number of layers of sheets is adjusted to suit the structural requirements;
3. Curing of FRP sheets applied; and
4. Finishing coat application on the FRP sheets

Some of the main features of the materials used and other aspects of using FRP sheets in repair and rehabilitation of structures are briefly described below.

Materials Used

Continuous Fiber Sheet

The unit weight of the sheet used is a direct reflection of the fiber content in the composite sheet, and indirectly the structural capacity of the sheet. It should be pointed out that resin and hardener penetration in very high density sheets is difficult, and therefore there is an optimum density of sheets for best structural performance, especially in cases when several layers of the sheets are used (CCC 1998).

Impregnation Resin

The principles that govern the choice of filler resins in FRP rods are also applicable for FRP sheets. Epoxy, vinyl ester, and acrylic resins are generally used for the purpose of impregnation. Structural properties of the hardened FRP sheet, such as strength and Young's modulus, rely upon the monolithic behavior of all the FRP layers. This can be ensured only if the resin completely impregnates the spaces between the fibers while it hardens. During application, the resin should have the required pot life and viscosity to hold the sheet in place. It may be pointed out that these properties are also related to the atmospheric temperature at the time of application, and therefore the resin should be carefully selected (CCC 1998).

Primer

Primer is applied to the concrete surface to ensure proper adhesion between the concrete surface and the FRP sheet and to pre-

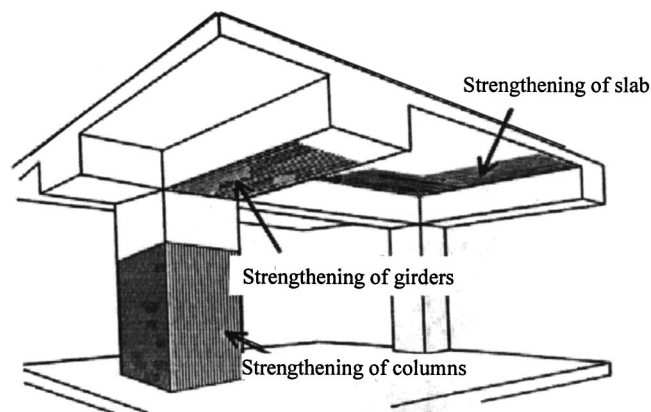


Fig. 17. Attachment of one-way reinforced fiber reinforced polymer sheets for increasing flexural capacity

vent formation of air bubbles at the interface, either during application of sheets or at a later time during curing.

Smoothing Agent

Unevenness in the concrete surface could become a source for accumulation of voids and wrinkles in the sheets. Such defects reduce the effectiveness of sheet application. Thus, in principle, all unevenness on the concrete surface should be removed before the FRP sheets are applied. Different methods, including application of a disc sander or a surface grinder, are used to suit the requirements. Sometimes, however, application of a “smoothing agent” may be sufficient for the purpose. The agent used should be selected keeping the required properties, temperature of application, etc., in mind. Epoxies have been reported to be used for this purpose.

Other Materials

Besides the materials outlined above, the application of FRP sheets is often preceded by impregnation of cracks in the concrete with an appropriate grout. Further, the FRP application may need to be provided with a finishing coat to improve its resistance to the environmental forces and prevent degradation on account of sunlight. Depending upon the environment of the structure, a fire-proofing coat may also be required.

Construction Method

Surface Preparation

This step should include rounding of the corners of members, abrading uneven surfaces using disc sanders, impregnation of cracks, and application of smoothing agents.

Application of Nonprestressed Fiber Reinforced Polymer Sheets

The surface is primed with resins before the sheets are pasted, and entrapped air is removed using rollers while coating with resins.

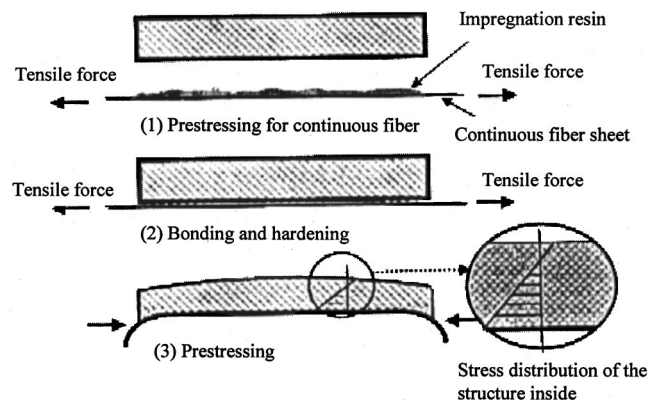


Fig. 19. Steps in application of prestressed fiber reinforced polymer sheets for repair and rehabilitation

In the case of multiple-layer application, the process is repeated after the application of each layer. Fig. 17 shows a schematic representation of application of FRP sheets for the reinforcement of different structural elements—columns, beams, and slabs. It may be noted that fibers in the sheets could be directed only in one direction or in two directions, again depending upon the desired results. For example, in the case of beams and columns, the fibers are aligned along the axes, whereas a two-way reinforcement is provided when using the sheets to reinforce a slab.

The different ways of attaching or wrapping the sheets (in a single- or multilayer application) are also shown in Fig. 18 for a beam-and-slab type of construction. It should be pointed out that it is often considered better to “wrap” the sheets, as shown in Fig. 18(d), to provide better structural behavior. However, it should be pointed out that FRP sheets can be more easily wrapped around such structures as chimneys or columns, where all the sides are usually accessible, rather than beams, where the compression face is usually not easily accessible. Another variation in applying the sheets is the use of preformed FRP plates around the structural members and making the joints of such plates monolithic subsequently. The space between the concrete and the FRP plates is filled or injected with cement or polymer mortar (Fujimura et al. 1998). In this case, one-way reinforced continuous FRP sheets are attached to the tensile face of a flexural member with fibers oriented along the member axis to provide increased flexural capacity (Kimura et al. 1997).

Application of Prestressed Continuous Fiber Reinforced Polymer Sheets

In the methods described above, the FRP sheets are simply bonded to the concrete surface using an impregnation resin. The effectiveness and reliability of such a sheet application has been found to be reduced on account of the facts that the FRP sheets do not participate in resisting dead load and that they play a part in resisting only live load. Further, such FRP sheets have a tendency to peel off even before the concrete member reached the ultimate state. To overcome the above problems, a method using pre-



Fig. 18. Schematic representation of different wrapping methods to increase flexural and shear capacity of RC members

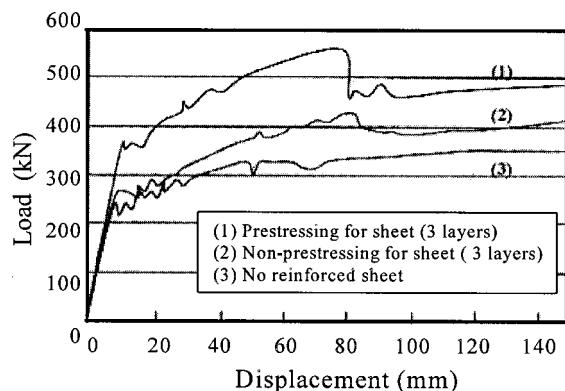


Fig. 20. Effectiveness of prestressed fiber reinforced polymer sheets in enhancing capacity of concrete members

stressed FRP sheets for bonding to the concrete surface has been developed and briefly described here (Wu et al. 2001). The principle of the method is illustrated in Fig. 19. The basic steps in the process are

1. Preparation of concrete surface in the usual manner (as for nonprestressed sheets);
2. Prestressing the continuous FRP sheet;
3. Applying the adhesive resin along the plane of adhesion and bonding to the concrete surface;
4. Curing and hardening of hardener with the tensile force held in place. At the ends, the reinforcement is carried out using a sheet fitted with steel plates, which are anchored to the concrete using bolts; and
5. Removing the externally applied load once sufficient hardening has been achieved, introducing a compressive “pre-stress” in the FRP sheets. Excess sheets at the ends are cut off.

Results of research work carried out to study the effectiveness of the application of prestressed FRP sheets in increasing the flexural capacity of RC members have been reported (Wu et al. 2001). Fig. 20 shows the results reported in the study. It was found that, besides toughness, the cracking load, yield load for the reinforcing bars, and the breaking loads are higher than those observed with simple (unprestressed) FRP sheet applications. It was also found that upon introduction of prestressing force, no peeling at the surface was observed, and that the failure was by rupture of the sheet.

Curing

During the hardening process of the hardeners (resins), especially in outdoor applications, it is important that the resins are protected from the action of rain, sand, dirt, etc. The temperature of the atmosphere also needs to be carefully monitored to ensure proper and adequate strength development in the resins (JRTRI 1996, 1997; JSCE 2001).

Finishing

After the resin has sufficiently hardened, the finishing coat may be applied with the object of ensuring long-term durability of the FRP sheet, fire protection, aesthetic appeal, etc. It may also be pointed out here that in order to ensure that the sheets do not debond from the concrete surface due to differential deformation during loading, the sheet is sometimes wrapped at the ends also, as shown in Fig. 18(d). This end protection provides an effect similar to that provided by the driving of anchors into the concrete in the case of use of steel plates for the strengthening, which

is far more cumbersome, especially in cases when the beams are heavily reinforced and it is difficult to find space for the anchors.

Design Methodology

General

The Japan Society of Civil Engineers (JSCE) and the American Concrete Institute (ACI) have published recommendations for the design and construction of concrete structures using FRP reinforcement (ACI 2001; JSCE 1997, 2001). These documents cover the requirements for materials used, their characteristic strength, and design recommendations for flexure and shear. The design for flexural members is based on the principles of equilibrium and compatibility and the constitutive laws of the materials. The recommended design methodology is based on the limit-state philosophy, and the design is checked for fatigue, creep, and serviceability criteria.

In addition to the application of FRP rods and tendons in the construction of new structures, FRP sheets have been extensively used in Japan in the retrofitting and upgrading of existing structures, especially after the Kobe earthquake in 1995. The FRP sheets have been used to increase the flexural and shear strength as well as ductility in many RC structures. Though methods such as jacketing with steel plates, etc., had been traditionally used for reinforcement, the development of this method has given the engineers a very useful option for rehabilitation of structures.

Design using Fiber Reinforced Polymer Rods or Tendons

Although only the details relating to design for flexure and shear have been briefly discussed in this paper, the JSCE also explains procedures for the design of prestressed concrete structures, design for fatigue, details for working out parameters like the flexural crack width, provisions for the cover thickness, and construction details—minimum concrete strength, handling, etc. (JSCE 1997, 2001).

The principles are similar to those used for normal RC design using steel reinforcement, and the flexural capacity of a section having FRP reinforcement can be calculated using the following assumptions:

- Strain in the concrete and the FRP reinforcement is proportional to the distance from the neutral axis (i.e., a plane section before loading remains plane after loading),
- The maximum usable compressive strain in concrete is assumed to be 0.003,
- The tensile strength of concrete is ignored,
- The tensile behavior of the FRP reinforcement is linearly elastic until failure, and
- A perfect bond exists between concrete and FRP reinforcement.

For flexural compressive failure, the compressive stress distribution in the concrete may be assumed to be identical to the rectangular compressive stress distribution (equivalent stress block). Concepts such as that of the reinforcement ratio (area of reinforcement in relation to the cross-sectional area of the member) can be used for FRP reinforced concrete members also. However, in determining the balanced reinforcement (when the tensile and compressive load-carrying capacities are the same), percentage, especially from the point of view of design, care needs to be taken to appropriately account for the brittle nature of the FRP composite failure.

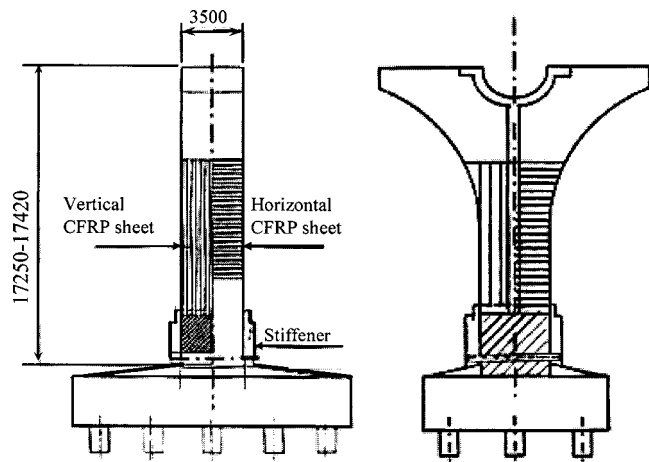


Fig. 21. Improvement in ductility of structure using fiber sheets in conjunction with steel plates

Now, the basic capacity of the section is determined either by the rupture of FRP due to tension or the crushing of concrete in compression. The implications of having the tensile reinforcement laid out in several layers have already been discussed in this paper.

Although the design procedure for the utilization of FRPs is still in its infancy, the following may be noted about the JSCE recommendations: (1) They ignore the presence of the FRP composites' presence in the compression zone; (2) modification factors are provided to account for the brittle failure of the FRPs; (3) issues like combined biaxial bending and axial forces are also addressed; and (4) the treatment of shear resistance is essentially an adaptation of the traditional method of adding individual contributions from concrete, shear reinforcement, and bent-up tendons.

Design Methodology for Upgrading using Fiber Reinforced Polymer Sheets

Fig. 21 shows a typical application of FRP sheets for the reinforcement of deteriorated concrete structures. It may be noted that the application is often carried out by using multiple layers of FRP sheets laid one after another, appropriately hardened together using a suitable hardener. Fig. 22 shows the variation of the strain and stress across the face of a beam reinforced with n layers of FRP sheets at the bottom. It can be seen that the resultant of all the layers is taken to be acting through one point and the thickness of the individual sheets has been neglected.

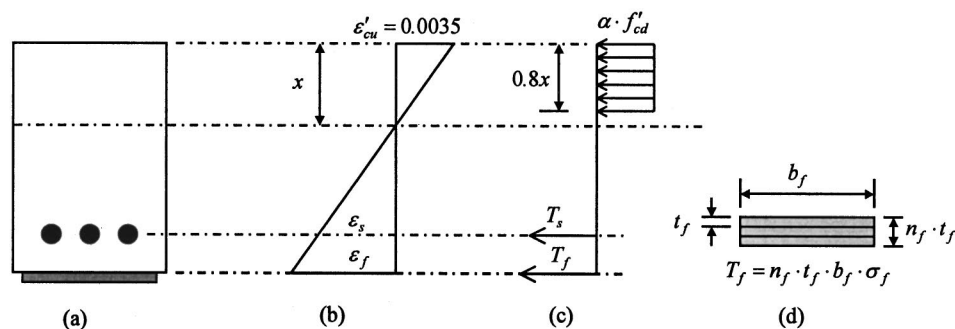


Fig. 22. Variation of strain and stress across beam face reinforced with n layers of CFRP sheets at bottom

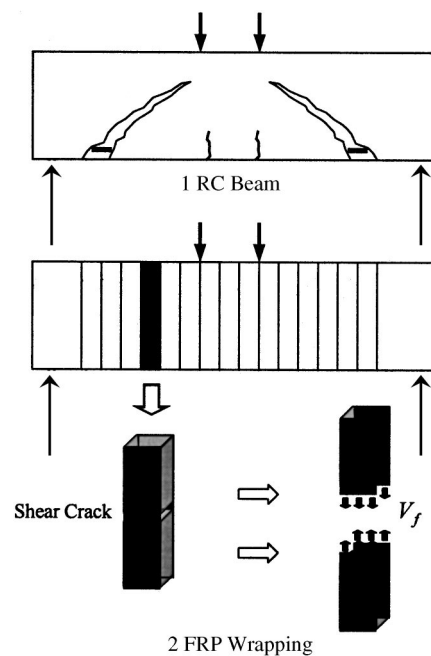


Fig. 23. Typical representation of truss model in study of contribution of continuous fiber reinforced polymer sheets to shear capacity of RC members

The sheets thus attached to concrete members contribute to (1) increasing the flexural and shear capacity of the section, and (2) increasing the ductility of the section (JSCE 2001; Katsuki et al. 2000; Nakai et al. 2000). As a result of the contributions to the flexural and shear capacities and the ductility of the members, members reinforced with FRP sheets also exhibit a larger deformation capacity compared with unreinforced members. Fig. 23 shows a schematic representation of the traditional truss model, which can be used to evaluate the contribution of FRP sheet(s) to the shear capacity of the RC member.

Examples of Application

In spite of their low modulus of elasticity and deformability at the ultimate load, FRP rods, grids, and sheets have been increasingly used as reinforcing material both in reinforced concrete, prestressed concrete, and repair and rehabilitation of concrete structures in the last decade. The nonmagnetic and noncorrosive nature of the FRPs also gives these materials an edge in specific appli-

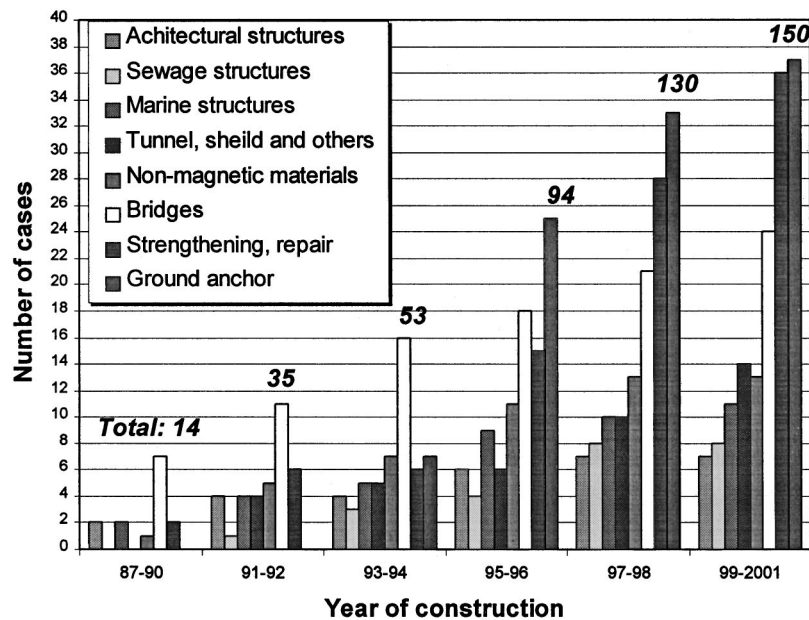


Fig. 24. Growing applications of fiber reinforced polymer composites in construction industry

cations. Fig. 24 gives a representation of some of the applications of the FRP reported in the literature. Though the list used is by no means exhaustive, it is clearly a pointer to the wide spectrum of possible applications of the FRP in the construction industry. In fact, the figure clearly shows that not only the number of applications show an increasing trend, but also that FRPs are being used in newer areas (ACC 1995, 1998).

General Application of Rods

Marine Concrete Structures

The noncorrosive nature of the FRPs makes them an excellent reinforcing material in the marine environment. One such application is shown in Fig. 25, where the Hexagonal Marine Structure constructed using FRP rods was used for tensioning of diagonals. Another application of FRP in marine structures is given in Fig. 26, where the rods were used in the marine slope stabilization. It may also be noted that FRP rods are also at times used in conjunction with other corrosion-resistant reinforcement, such as epoxy-coated bars.

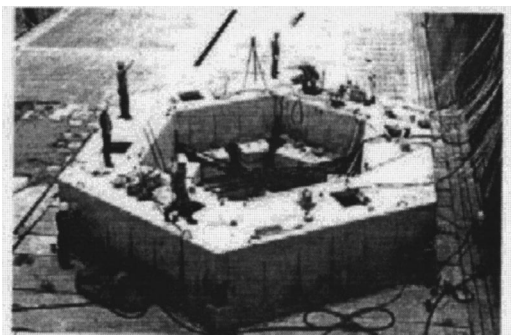


Fig. 25. Hexagonal marine structure

Bridges (PC)

FRP composites have been used extensively in bridges, especially in prestressed concrete construction. The first actual structure to be constructed using continuous fibers as tendons was the Ulenberg Bridge in Germany, built in 1986. The bridge is a continuous slab bridge and its tendons are multiple strands made of 7.5-mm-diameter FRP. This was preceded by a small experimental bridge spanning only about 6 m, also in Germany, in the early 1980s.

A pedestrian bridge (Marienfede Bridge) in Berlin Park, the Ludwigshafen Bridge, and the Lunen'sche-Gasse Bridge are examples of FRP applications in Germany (ACI 2001). Notsch Bridge, constructed in 1991 in Austria, also used GFRP tendons as prestressing tendons. A precast posttensioned bridge of 9 m span and 5.2 m width was erected at a cement plant in Rapid City, South Dakota, in 1992. Cables of three different materials were used to prestress the slab. Thirty GFRP cables were used to pre-

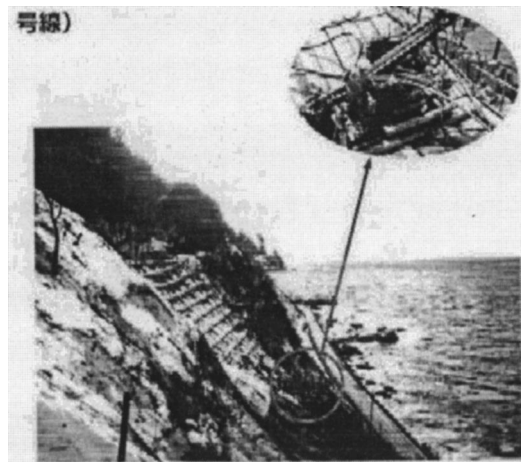


Fig. 26. Application of fiber reinforced polymer in marine slope stabilization

FRP Rods

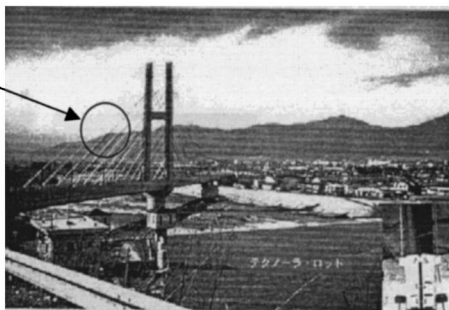


Fig. 27. Use of fiber reinforced polymer composites as cables in cable-stayed bridge

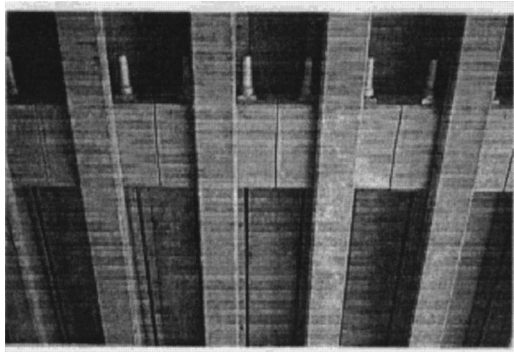


Fig. 28. Use of fiber reinforced polymer rods as external cables in rehabilitation project

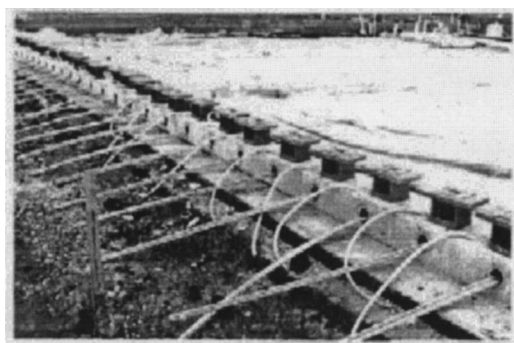


Fig. 29. Use of nonmagnetic fiber reinforced polymer reinforcement in compass alignment apron area of Haneda airport, Japan

stress one-third of the length of the slab, 30 FRP cables were used to prestress the next third of the length, and steel cable prestressing was adopted for the remaining length (ACI 2001). FRP bars were also used in the U.S. for the deck construction of Pierce Street Bridge in Lima, Ohio, in 1999 and the redecking of Dayton, Ohio's Salem Avenue Bridge in 1999 (ACI 2001).

The "Birdie Bridge" is one of the first applications of continuous FRP material in bridge construction in Japan. In this bridge, a total of 16 cables each consisting of 8 AFRP flats were used as tendons. Besides being used as reinforcing materials within concrete, the FRP rods have also been used as cables in suspension and cable-stayed bridges, such as is shown in Fig. 27.

Besides examples of use in new construction, as discussed above, FRP rods have also been used in the repair and rehabilitation works. Fig. 28 shows an example where such rods were used



Fig. 30. Use of fiber reinforced polymer in conjunction with conventional reinforcement in parking area

as external cables in the rehabilitation work.

Transport Facilities and Tunnel Linings

FRP rods have also found application as reinforcing materials in applications where the magnetic nature of the conventional steel reinforcement becomes a bottleneck. Such examples include the experimental facility for the high-speed linear motor-rails. Fig. 29 shows the application of FRP tendons in the construction of the compass alignment apron area of the Haneda airport in Tokyo, Japan, where the presence of magnetic materials in the slab interfere with the operation of devices. Fig. 30 shows an example where FRP rods are used in conjunction with conventional reinforcement in a parking area. The nonmagnetic FRP reinforcement is used only in those areas where cars are designated to park and magnets are used for operational purposes.

FRP grids are becoming increasingly popular as a replacement of steel meshes used in the shotcreting in tunnels. Fig. 31 shows an example of the application of NEFMAC fibers in an underground oil storage facility in Kagoshima, in Japan.

Building Structures

Glass FRP bars have also been used for the beams and walls in a University building and the construction of the MRI unit of a hospital building in San Antonio (ACI 2001). Fig. 32 shows the application of a 2D FRP composite in building construction. Several applications of FRP using carbon fibers in curtain walls of buildings have also been reported.

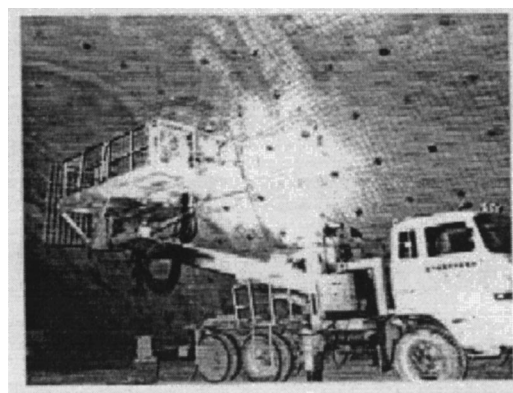


Fig. 31. Use of fiber reinforced polymer grids in shotcreting in underground oil storage facility in Kagoshima, Japan

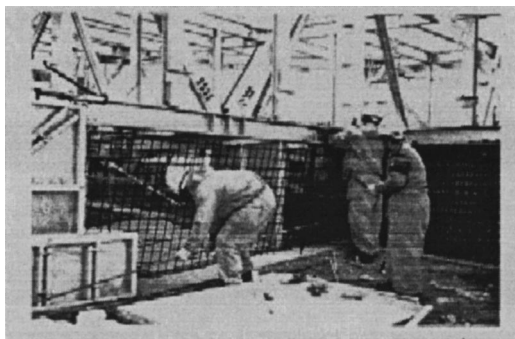


Fig. 32. Use of two-dimensional fiber reinforced polymer reinforcement in building construction

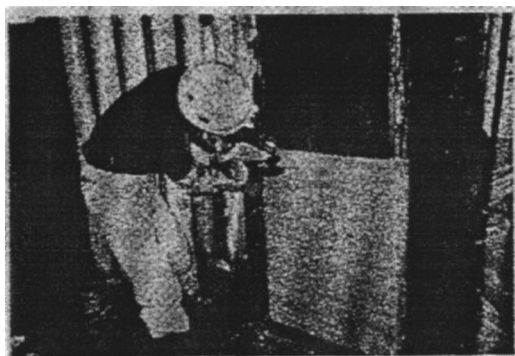


Fig. 33. Use of fiber reinforced polymer sheets to strengthen RC columns

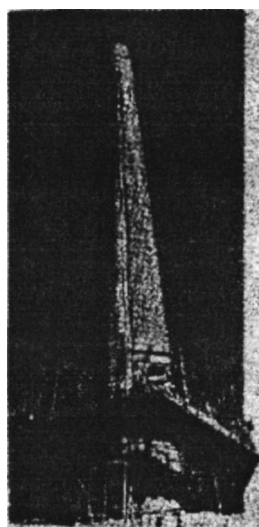


Fig. 34. Chimney repair using fiber reinforced polymer sheets

Applications of Continuous Sheets for Repair and Rehabilitation

Sheets made from carbon and aramid fibers have been found to be especially useful in retrofitting works to improve the seismic performance of buildings and bridges, rehabilitation of deterioration, and damaged structures. Projects have involved the use of FRP sheets for several hundred to several thousand square meters of area. Some of the examples are given in this section to show the diversity of the application.

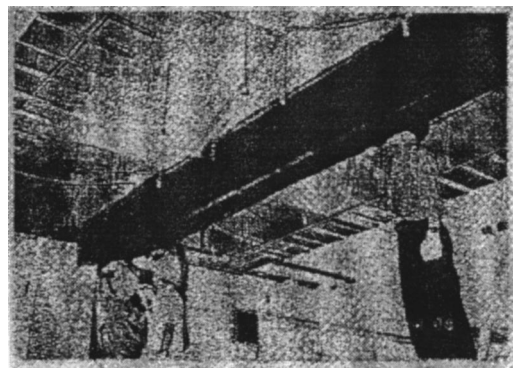


Fig. 35. Strengthening of beams and slabs in school building

For example, a change in the design wheel loads, in Japan in 1993, necessitated retrofitting a large number of highway bridges. Carbon FRP sheets have been used to strengthen flexural and shear capacities of structures including highway piers. Reference may be made to Fig. 21 for the application of continuous FRP sheets in conjunction with steel plates to reinforce piers. Fig. 33 shows another example of the use of continuous FRP sheets for the reinforcement of columns. FRP sheets have also been applied to repair the deterioration of jetties and other structures on account of chloride-induced corrosion, tunnel linings to strengthen the concrete walls and help resisting the ground pressure and repair of cracks in tunnels. Fig. 34 shows the example of application of FRP sheets in the repair works of a chimney, damaged on account of reinforcement corrosion, exhaust fumes, and temperature-related distress.

Fig. 35 shows an example of the use of FRP sheets for the strengthening of the floor slabs and beams in a school slab. It can be seen that FRP sheets are used on both the top and bottom surfaces of the slab to improve the structural integrity and prevent further propagation of cracks. It may be noted that use of FRP sheets as discussed here allows the designer the flexibility to use different number of layers in different directions in cases when there is a differential requirement for strengthening in the two directions.

Concluding Remarks

An attempt has been made in this paper to briefly discuss the different aspects related to the utilization of continuous FRP composites as a reinforcing material in concrete construction. Given the diversity involved in the issues, this paper has largely focused on providing information only on the core engineering aspects—the structural and durability characteristics of FRP composite, research work related to the utilization of FRPs as a reinforcing material, and applications of FRP composites, including sheets. Several examples of utilization in the field have been included to demonstrate the range of possible applications. Since issues related to structural design were not the primary focus of this paper, the issue has been only briefly discussed. Other issues, such as the manufacture of FRPs, have not been discussed in any length.

As can be clearly seen from the information presented in this paper, there is clearly a growing need and acceptability of FRP composites in the field of concrete engineering, and FRP composites are being used in a growing number of applications. Thus, the coming years are likely to see a rapid increase in the development of appropriate test methods and specifications, design methods in

the field of reinforced and prestressed concrete construction, and increased field applications. More research effort is called for in areas such as making the FRPs cost-effective and developing appropriate construction technology to implement designs in the field.

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Appendix I. Test Methods for Continuous Fiber Reinforcing Materials (JSCE 1997)

1. Test method for tensile properties of continuous fiber reinforcing materials (JSCE-E 531-1995)
2. Test method for flexural tensile properties of continuous fiber reinforcing materials (JSCE-E 532-1995)
3. Test method for creep failure of continuous fiber reinforcing materials (JSCE-E 533-1995)
4. Test method for long-term relaxation of continuous fiber reinforcing materials (JSCE-E 534-1995)
5. Test method for tensile fatigue of continuous fiber reinforcing materials (JSCE-E 535-1995)
6. Test method for coefficient of thermal expansion of continuous fiber reinforcing materials by thermal-mechanical analysis (JSCE-E 536-1995)
7. Test method for performance of anchorages and couplers in prestressed concrete using continuous fiber reinforcing materials (JSCE-E 537-1995)
8. Test method for alkali resistance of continuous fiber reinforcing materials (JSCE-E 538-1995)
9. Test method for bond strength of continuous fiber reinforcing materials (JSCE-E 539-1995)
10. Test method for shear properties of continuous fiber reinforcing materials (JSCE-E 540-1995)

Appendix II. Test Methods for Continuous Fiber Sheets (JSCE 2001)

1. Test method for tensile properties of continuous fiber sheets (JSCE-E 541-2000)
2. Test method for overlap splice strength of continuous fiber sheets (JSCE-E 542-2000)
3. Test method for bond properties of continuous fiber sheets to concrete (JSCE-E 543-2000)
4. Test method for bond properties of continuous fiber sheets to steel plate (JSCE-E 544-2000)
5. Test method for direct pull-off strength of continuous fiber sheets with concrete (JSCE-E 545-2000)
6. Test method for tensile fatigue strength of continuous fiber sheets (JSCE-E 546-2000)
7. Test method for accelerated artificial exposure of continuous fiber sheets (JSCE-E 547-2000)
8. Test method for freeze-thaw resistance of continuous fiber sheets (JSCE-E 548-2000)

9. Test method for water, acid, and alkali resistance of continuous fiber reinforcing materials (JSCE-E 549-1995)
10. Test method for flexural tensile strength of continuous fiber reinforcing materials (DRAFT)
11. Test method for surface incombustibility of protective materials for continuous fiber reinforcing materials (DRAFT)

Notation

The following symbols are used in this paper:

- E_{FRP} = modulus of elasticity of FRP composite, e.g., rod, etc.;
- E_1 = modulus of elasticity of fiber in composite;
- E_2 = modulus of elasticity of matrix phase in composite;
- f_{ck} = characteristic compressive strength of concrete in N/mm²;
- n = number of layers of FRP sheets used to reinforce member;
- V_f = volume fraction of fibers in FRP composite;
- α = modification factor for estimation of bond strength of FRP composite with concrete;
- γ_c = constant = 1.3;
- σ_{FRP} = stress in FRP composite, e.g., rod;
- σ_1 = stress carried by fiber phase in composite;
- σ_2 = stress carried by matrix phase in composite; and
- $\square_u$ = tensile strength of FRP composite.

References

- Advanced Composite Cables. (1995). *Applications and construction using new materials*, ACC Club, Tokyo (in Japanese).
- Advanced Composite Cables. (1998). *Applications and construction using new materials*, Vol. 2, ACC Club, Tokyo (in Japanese).
- Agarwal, B. D. and Broutman, L. J. (1990). *Analysis and performance of fiber composites*, Wiley, New York, 44–45.
- American Concrete Institute. (2001). "Guide for the design and construction of externally bonded FRP systems for strengthening concrete structures." *Rep. by ACI Committee 440*.
- CCC. (1998). *Fiber-reinforced composites for construction*, CMC Co., T. Nakatsuji, ed. (in Japanese).
- Fujimura, M., et al. (1998). "Strength and behavior of RC column strengthened by FRP plates." *Proc. Japan Concrete Institute*, 20(3), 1201–1206 (in Japanese).
- Hodhod, H. A. A. (1992). "Employment of constituent properties in evaluation and interpretation of FRP rods mechanical behavior." DEng. thesis, Univ. of Tokyo.
- Iwahashi, T., et al. (1996). "Studies on the repairing and strengthening of existing RC column." *Proc. JCI*, 18(2), 1499–1504 (in Japanese).
- Japan Concrete Institute. (1997). *Report of technical committee on continuous fiber-reinforced concrete* (in Japanese).
- Japan Concrete Institute. (1998). "Technical report on continuous fiber-reinforced concrete." *JCI TC952*.
- Japan Railway Technical Research Institute. (1996). *Recommendation in the design and practice for seismic strengthening the pier of railway bridges by carbon fiber sheet* (in Japanese).
- Japan Railway Technical Research Institute. (1997). *Recommendation in the design and practice for seismic strengthening the pier of railway bridges by aramid fiber sheet* (in Japanese).
- Japan Society of Civil Engineers. (1997). "Recommendations for design and construction of concrete structures using continuous fiber reinforcing materials." *Concrete Engineering Series*, No. 23.
- Japan Society of Civil Engineers. (2001). "Recommendations for upgrading of concrete structures with use of continuous fiber sheets." *Concrete Engineering Series*, No. 41.

- Jesse, F., and Curbach, M. (2001). "The present and the future of textile reinforced concrete." *Proc., 5th Int. Conf. on Fiber-Reinforced Plastics for Reinforced Concrete Structures*, C. J. Burgoyne, ed., Cambridge, U.K.
- Katsuki, F. (1997). "Evaluation of alkali resistance of fiber reinforcement for concrete." Thesis submitted to the Univ. of Tokyo for D. Engrg. degree (in Japanese).
- Katsuki, F., et al. (2000). "Ductility evaluation of RC member retrofitted with continuous fiber sheet." *Proc. Jpn. Concr. Inst.*, 22(3), 1537–1542 (in Japanese).
- Katz, A., Bank, L. C., and Puterman, M. (2001). "Durability of FRP rebars after four years of exposure." *Proc., 5th Int. Conf. on Fiber-Reinforced Plastics for Reinforced Concrete Structures*, C. J. Burgoyne, ed., Cambridge, U.K.
- Kimura, K., et al. (1997). "Flexural behaviors reinforced concrete members with FRP plates." *Proc. Jpn. Concr. Inst.*, 19(2), 267–272 (in Japanese).
- Kobayashi, K., Uomoto, T., and Cho, R. (1987). "Structure of prestressed concrete using one-dimensional fiber composites as reinforcement." *J. Jpn. Soc. Compos. Mater.*, 13(5).
- Mallick, P. K. (1988). *Fiber-reinforced composites—materials, manufacturing and design*, Marcel Dekker, New York.
- Nakai, H., et al. (2000). "Evaluation of shear capacity of linear members retrofitted by continuous fiber sheets." *Proc. Jpn. Concr. Inst.*, 22, 493–498 (in Japanese).
- Nishimura, T., Uomoto, T., Katsuki, F., and Kamiyoshi, M. (1996). "Development of alkali resistant AGFRP rods." *Proc., 51st Annual Meeting of the JSCE (Part V)* (in Japanese).
- Piggott, M. R., and Harris, B. (1980). "Compression strength of carbon, glass and Kevlar-49 fiber reinforced polyester resins." *J. Mater. Sci.*, 15, 2523.
- Saadatmanesh, H., and Tannous, F. E. (1999). "Relaxation, creep and fatigue behavior of carbon fiber reinforced plastic tendons." *ACI Mater. J.*, March–April, 143–153.
- Taerwe, L., Matthys, S., Pilakouts, K., and Guadagnini. (2001). "European activities on the use of FRP reinforcement, fib Task Group 9.3 and the ConFiberCrete network." *Proc., 5th Int. Conf. on Fiber-Reinforced Plastics for Reinforced Concrete Structures*, C. J. Burgoyne, ed., Cambridge, U.K.
- Toritani, T., et al. (1997). "Experimental study on aseismic strengthening of existing reinforced concrete columns, Part 2, discussion on experimental results." *Summaries of Technical Papers of Annual Meeting, AIJ*, 23330, 659–660 (in Japanese).
- Uomoto, T., Tajima, T., and Nishimura, T. (1992). "Relation between acoustic emission and absorbed energy of reinforced concrete and fiber-reinforced plastic." *Progress in Acoustic Emission VI*, Japanese Society for NDI.
- Uomoto, T., and Nishimura, T. (1994). "Static strength and elastic modulus of FRP rods for concrete reinforcement." *Concrete Library of the Japan Society of Civil Engineers*, No. 23.
- Uomoto, T., Ohga, H., Nishimura, T., and Yamaguchi, T. (1995). "Fatigue strength of FRP rods for concrete reinforcement." *Proc., EASEC*, Y. C. Loo, ed.
- Uomoto, T., Nishimura, T., Ohga, H., Katsuki, F., Yamaguchi, T., and Kato, Y. (1998). "Strength and durability of FRP rods for prestressed concrete tendons." *Rep. of the Institute of Industrial Science, Univ. of Tokyo*, 39(2), (Serial 244) (in Japanese).
- Uomoto, T. (2001). "Durability considerations for FRP reinforcements." *Proc., 5th Int. Conf. on Fiber-Reinforced Plastics for Reinforced Concrete Structures*, C. J. Burgoyne, ed., Cambridge, U.K.
- Waaser, E., and Wolf, R. (1986). *Ein neuer werkstoff für spannbeton HLV Hochleistungs-Verbundstab aus glasfasern*, Beton H 7 (in German).
- Wu, Z., et al. (2001). "Experimental study on strengthening method of PC flexural structures with externally bonded prestressed fiber sheet." *Proc. Jpn. Concr. Inst.*, 23(1), 1099–1104 (in Japanese).
- Yagishita, K., et al. (1997). "Experimental study on aseismic strengthening of existing reinforced concrete columns, Part 1—outline of experiment." *Summaries of Technical Papers of Annual Meeting, AIJ*, 23329, 657–658 (in Japanese).
- Yamaguchi, T. (1998). "Deterioration due to ultraviolet radiation and the creep failure of FRP rods." DEngg. thesis, Univ. of Tokyo (in Japanese).